

7.2.5 Software Validation Process

NOTE — The Software Validation Process confirms that software products fulfill their intended use in the target operational environment. Validation may involve testing, analysis, simulation, modeling, and operational evaluation. Validation is typically performed after verification is complete, but may be conducted concurrently for incremental or iterative life cycles.

7.2.5.1 Purpose

The purpose of the Software Validation Process is to **confirm that the requirements for a specific intended use of the software work product are fulfilled**, demonstrating that the software meets user needs and intended operational scenarios.

7.2.5.2 Outcomes

As a result of successful implementation of the Software Validation Process:

- a) a validation strategy is developed and implemented;
 - b) criteria for validation of all required work products are identified;
 - c) required validation activities are performed;
 - d) problems are identified and recorded;
 - e) evidence is provided that the software work products, as developed, are suitable for their intended use; and
 - f) results of the validation activities are made available to the customer and other involved parties.
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7.2.5.3 Activities and Tasks

The project shall implement the following activities and tasks in accordance with applicable organizational policies and procedures with respect to the Software Validation Process.

7.2.5.3.1 Process Implementation

This activity consists of the following tasks:

7.2.5.3.1.1 Validation Need and Independence Determination

A determination shall be made whether the project warrants a validation effort and the degree of **organizational independence** required for that effort, based on criticality, risk, and contractual or regulatory requirements.

7.2.5.3.1.2 Validation Process Establishment

If the project warrants a validation effort, a validation process shall be established to validate the system or software product. Validation tasks defined in 7.2.5.3.2, including associated methods, techniques, and tools, shall be selected based on the target environment and intended use.

7.2.5.3.1.3 Independent Validation Organization

If the project warrants an independent validation effort, a qualified organization responsible for conducting the effort shall be selected. This organization shall be assured of the **independence and authority** to perform the validation tasks objectively.

7.2.5.3.1.4 Validation Planning

A **Validation Plan** shall be developed and documented. The plan shall include, but is not limited to, the following:

- a) items subject to validation;
 - b) validation tasks to be performed;
 - c) resources, responsibilities, and schedule for validation;
 - d) procedures for forwarding validation reports to the acquirer and other involved parties;
 - e) acceptance criteria and success metrics for each validation activity;
 - f) test environments, tools, and data required to conduct validation.
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7.2.5.3.1.5 Validation Execution and Reporting

The validation plan shall be implemented. Problems and non-conformances detected by the validation effort shall be entered into the **Software Problem Resolution Process (7.2.8)**. All problems and non-conformances shall be

resolved. Results of the validation activities shall be documented and made available to the acquirer and other involved organizations.

7.2.5.3.2 Validation

NOTE — Other means besides testing (such as analysis, modeling, simulation, inspections, or operational demonstrations) may be employed for validation where appropriate.

This activity consists of the following tasks:

7.2.5.3.2.1 Test Preparation

Prepare selected test requirements, test cases, and test specifications for analyzing validation results. These shall be based on intended use cases, operational scenarios, and end-user needs.

7.2.5.3.2.2 Test Case Alignment with Intended Use

Ensure that validation test requirements, test cases, and test specifications **reflect the particular requirements for the specific intended use** of the software product, including realistic operational environments and workflows.

7.2.5.3.2.3 Validation Test Execution

Conduct the validation tests prepared under 7.2.5.3.2.1 and 7.2.5.3.2.2, including:

- a) testing with **stress, boundary, and singular inputs**;
 - b) testing the software product for its ability to **isolate and minimize the effect of errors**, including graceful degradation under stress, boundary, or singular conditions, and appropriate operator assistance requests;
 - c) testing that **representative users** can successfully achieve their intended tasks using the software product under realistic operational conditions.
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7.2.5.3.2.4 Validation of Intended Use

Validate that the software product satisfies its intended use by evaluating test outcomes, operational scenarios, and user feedback against validation criteria and

acceptance metrics.

7.2.5.3.2.5 Target Environment Validation

Where appropriate, test the software product in **selected areas of the target environment** to confirm its suitability for deployment and operation in real or representative conditions.

7.2.5.4 Deliverables

The following deliverables shall be produced during this process:

- Software Validation Plan;
 - Validation test requirements, test cases, and specifications;
 - Validation test procedures and reports;
 - Records of problems, non-conformances, and their resolution;
 - Validation summary report and acceptance evidence;
 - Operational test results and user evaluation feedback;
 - Independent validation reports (if applicable).
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7.2.5.5 Audit and Configuration Control

All validation plans, test procedures, results, and reports shall be placed under appropriate **configuration management**.

Audit evidence shall be retained in accordance with organizational and regulatory policies.

Validation evidence shall be protected to ensure integrity, and baselines shall be updated upon resolution of issues and completion of validation activities.

Cross-References

This process interacts with the following:

- **Software Requirements Analysis (7.1.2)** — provides requirements and intended use cases for validation.
- **Software Architectural Design (7.1.3), Detailed Design (7.1.4), and Construction (7.1.5)** — provide work products subject to validation.
- **Software Integration (7.1.6) and Software Qualification Testing (7.1.7)** — provide integrated products and test evidence for validation.

- **Software Quality Assurance (7.2.3)** — coordinates validation activities for compliance with plans.
- **Software Verification (7.2.4)** — provides verification evidence used to support validation planning.
- **Software Review (7.2.6)** and **Software Audit (7.2.7)** — provide additional oversight and independent evaluations.
- **Software Problem Resolution (7.2.8)** — receives problems and non-conformances found during validation for resolution tracking.